

A CONVERSION BENCHMARK

LaTeX $\rightarrow$ Typst

What three deterministic engines and two AI models can — and cannot — do with real documents.

48 samples

15 categories

3 engines

2 AI models

SCOPE

What We Tested

Three deterministic engines — **Pandoc**, **Tylax**, **TypeTeX** — were run across all categories — **Gemini** and **GPT**, manually prompted with no API access — were run across all categories. Gemini is scored against an absolute, per-candidate rubric rather than ranked relative to other candidates.

THE 15 CATEGORIES

- Prose
- Tables
- CVs
- Beamer Decks
- Equations
- Algorithms
- Graphics – TikZ / PGFPlots / Posters

METHOD

Two Layers of Scoring

LAYER 1

The Rubric

Reads the generated Typst source directly. Does it compile? Is it clean? Did the text, structure, numbering, and styling survive translation?

LAYER 2

Visual

Rendered output. Does it look like the reference? Are there any errors in the rendering?

The two layers usually agree. When they don't — see the beamer case later — the visual layer wins.

METHOD

The Six-Point Rubric

01

Compiles

Does the Typst source compile cleanly, with no fatal errors?

02

Clean Source

Is the output free of leaked, unresolved LaTeX commands?

04

Structural Elements

Are headings, lists, tables, and equations real Typst structures — not flattened text?

05

Numbering Correctness

Do figures, tables, and equations have correct Typst numbering and references?

A failed compile scores zero on all six.

METHOD

Visual Alignment: How It’s Measured

IoU — Text Position

Text block bounding boxes, normalized by page size and matched by reading order, scored by intersection-over-union.

Exact ≥ 0.70
Minor $0.40\text{--}0.70$
Major < 0.40

SSIM — Graphics

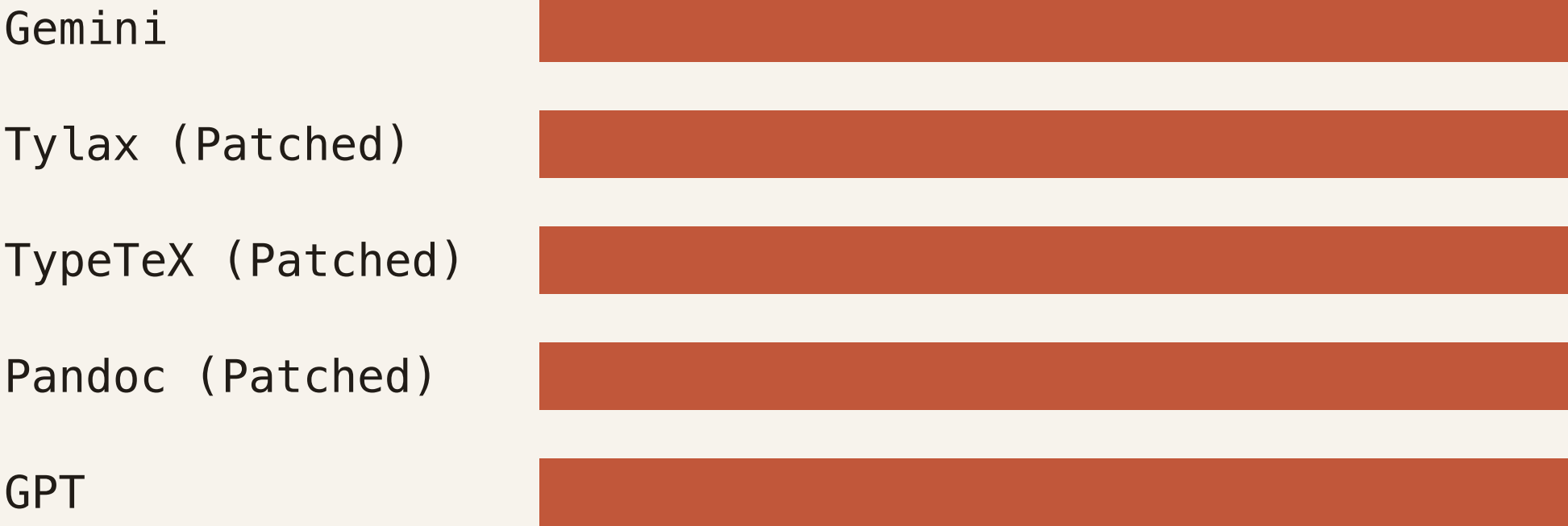
For TikZ, PGFPlots, and posters — no reliable text layer — rendered to compared structurally.

Exact ≥ 0.95
Minor $0.75\text{--}0.95$
Major < 0.75

THE NUMBERS

Headline Scores: Fair Subset

8 categories tested by both AI models and engines — average score out of 6.0



THE NUMBERS

Headline Scores: Full Corpus

15 categories, engines only — as-tested baseline vs. patched ceiling

Pandoc



TypeTeX



Tylax



As-tested Patched ceiling

FINDING 01

AI vs. Engines: Close, Not a Blowout

5.88 5.50

Gemini

Tylax (Patched)

On the 8 categories both were tested on, Gemini leads — but Tylax, Type behind. This comparison is fair only within these 8 categories: AI models v posters, PGFPlots, TikZ) due to manual-prompting constraints, so no full-

FINDING 02

GPT’s Compile Reliability

2 / 8

GPT samples failed to compile

0 / 8

Gemini samples failed

Both GPT failures scored 0/6. Gemini compiled every sample; its only mis doubled quotation marks — not a hard failure.

IN PRACTICE

When It Works, It Really Works

Recent advancements in large language models have demonstrated unprecedented capabilities across diverse natural language processing tasks. This rapid progress challenges traditional paradigms of linguistic analysis and demands robust evaluation frameworks to systematically assess model reasoning, factuality, and potential biases in zero-shot contexts.	Recent advancements in large language models have demonstrated unprecedented capabilities across diverse natural language processing tasks. This rapid progress challenges traditional paradigms of linguistic analysis and demands robust evaluation frameworks to systematically assess model reasoning, factuality, and potential biases in zero-shot contexts.
Reference	Gemini Match: 1.00 Pos: 0.681
Recent advancements in large language models have demonstrated unprecedented capabilities across diverse natural language processing tasks. This rapid progress challenges traditional paradigms of linguistic analysis and demands robust evaluation frameworks to systematically assess model reasoning, factuality, and potential biases in zero-shot contexts.	Recent advancements in large language models have demonstrated unprecedented capabilities across diverse natural language processing tasks. This rapid progress challenges traditional paradigms of linguistic analysis and demands robust evaluation frameworks to systematically assess model reasoning, factuality, and potential biases in zero-shot contexts.
Pandoc Match: 1.00 Pos: 0.738	Tylax Match: 1.00 Pos: 0.738

Sample **prose_easy** — every candidate, deterministic or AI, reproduces the reference and scores range 0.68–0.74. Gaps only start to show up as documents get structurally harder.

FINDING 03

TypeTeX vs. Pandoc at the Patched Ceiling

5.44

TypeTeX, patched ceiling

5.40

Pandoc, patched ceiling

97

TypeTeX

With the `\hrule` patch applied, TypeTeX compiles 47 of 48 samples — engine. What keeps it from pulling ahead is clean-source rate: **68.8%** vs. embeds raw LaTeX math inside `#mitex(...)` calls, which the rubric could approach, not a transcription bug.

FINDING 04

Tylax: the Structural Outlier

4.74

Tylax, patched ceiling

~5.4

TypeTeX & Pandoc, patched ceiling

Even after patching, Tylax trails the other two engines. It shows a recurring — across multiple unrelated categories, pointing to a broader structural w competitiveness (5.50) doesn't carry over to the full corpus, where CV- an

FINDING 05

Two Bugs, Found and Patched

Reserved-character leakage – Tylax only

An unescaped @ in plain text — e.g. an email address — breaks compilation, since Typst reserves @ for label references. Confirmed limited to 6 CV samples; patched to \@, which immediately resolved the crashes.

Macros

\hrule
comp
Patch

Grace Hopper Rear Admiral, US Navy ghopper@navy.mil Arlington, VA 1952 Invented the first operational compiler (A-0 System). 1959 Served as technical consultant on the CODASYL committee, leading to the creation of COBOL.	@X r@ Grace Hopper ghopper@navy.mil Rear Admiral, US Navy Arlington, VA @l X@ 1952 Invented the first operational compiler (A-0 System). 1959 Served as technical consultant on the CODASYL committee, leading to the creation of COBOL. Reference Tylax (patched) Match: 1.00 Pos: 0.000
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FINDING 06

TypeTeX’s Equation Pipeline

“A confirmed software defect in the filter itself — a traversal

TypeTeX’s math pipeline, run through its Lua filter, initially crashed outright patched at the engine level for this run, meaning its equations now compile an artificial crash.

FINDING 07

Beamer: Where the Two Layers Disagree

The `beamer_hard` sample — a multi-frame deck with `\titlepage`, `\block`, and `\section` commands — is a good test case for conversion engines: none preserve beamer’s frame/block hierarchy. Pandoc and TypeTeX both fail to generate a title page reference. Tylax’s conversion completes without error, but the generated PDF does not compile.

RUBRIC SAYS 6 / 6 — pass on Pandoc & TypeTeX — the missing title page isn’t enough prose to trip text-completeness	VISUAL LAYER Content zero content here
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The visual layer is the one reflecting what a reader actually sees.

IN PRACTICE

Content vs. Position: A Table Example

Taxonomy		Metrics	
Class	Order	Avg. Weight (g)	Avg. Length (cm)
Aves	Passeriformes	45.2	15.4
	Strigiformes	1200.5	45.0
Mammalia	Rodentia	25.4	10.2

Reference

Tylax | Ma

Sample **tables_complex_hard**, Tylax. The text is fully matched (**1.00**) — every word lar
reference, so position IoU is a moderate **0.537**. Content and position are graded separat

IN PRACTICE

Content vs. Position: An Equation Exam

The Taylor series expansion of a real or complex-valued function $f(x)$ about the point $x = a$ is given by: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \tag{1}$	The Taylor se given by:
Reference	Pandoc

Sample **eq_simple_hard**, Pandoc. Pandoc outputs Unicode math italics instead of styled reference words (**0.83**). But the equation lands in a visibly different vertical position, so l

THE NUMBERS

Full-Corpus Alignment Averages

All 48 samples — compile failures counted as 0.0

	TEXT CONTENT MATCH
Pandoc	0.679
Tylax	0.586
TypeTeX	0.691

Near-zero graphics SSIM is expected: none of the three engines can translate TikZ/PGFPLOT. nonzero SSIM comes from partial TikZ output, dominated by one sample (tikz_simple_m

CAVEATS

What the Visual Metric Misses

- 01 Subtle font and kerning differences — unless they shift a whole block.
- 02 Color mismatches — SSIM is computed in grayscale, to focus on struc
- 03 The actual reason for a mismatch — an overflowing table just reads as
- 04 Block-chunking artifacts — a reference emitting one whole-table block mismatch even when the rendered text is visually identical.

CLOSING

Where Things Stand

- On matched ground, Gemini leads — but Tylax, TypeTeX, and Pandoc
- At the patched ceiling, TypeTeX and Pandoc are nearly tied; Tylax rem
- Two specific engine bugs — the @ escape and the \hrule mistranslat
- Beamer decks and math-heavy pipelines remain the hardest cases for
- Rubric and visual-alignment scoring don't always agree — when they c